What We Can Compute — 11 NVIDIA NIMs Live

DiffDock v2.2	— Molecular docking (drug → protein binding prediction)
AlphaFold2	— Protein structure prediction from sequence
AlphaFold2-Multimer	— Protein complex structures (multi-chain)
ESMfold	— Fast protein folding (no MSA needed, seconds)
RFdiffusion	— De novo protein binder design
ProteinMPNN	— Sequence optimization for designed proteins
GenMol	— AI molecule generation for target pockets
MolMIM	— Molecule optimization with QED scoring
MSA Search	— ColabFold sequence alignment
ESM-2	— Protein embeddings (1280-dim structural fingerprints)
RDKit	— Drug-likeness, BBB, CNS MPO, PAINS screening

All accessible via REST API + MCP Server for AI agent integration

Computational Services We Offer

1 Target Analysis

Score any gene/protein across 5 evidence dimensions from 14,176 claims

2 Virtual Screening

Dock your compound library against any SMA target (DiffDock v2.2)

3 Protein Binder Design

Design novel proteins that bind ROCK2/LIMK1/MAPK14 (RFdiffusion + ProteinMPNN)

4 Drug Discovery

Generate + filter + dock novel molecules for your target (GenMol + RDKit + DiffDock)

5 Literature Synthesis

Cross-paper analysis across 6,535 sources — find connections no single paper reveals

6 Hypothesis Generation

AI-generated testable predictions with experiment proposals

All open source. Results delivered as structured data, ready for your analysis.

What’s Coming Next

Research Roadmap — Next 4 Weeks

Week 1

Cross-disease transcriptomics

20 ALS datasets × 24-gene panel — find shared and divergent expression signatures

Week 2

Extended DiffDock campaign

63 targets × top drug candidates — full binding landscape across the SMA target space

Week 3

Protein binder design

Novel proteins targeting ROCK2 / MAPK14 / LIMK1 (RFdiffusion + ProteinMPNN pipeline)

All computation runs autonomously. Results feed back into the platform daily.

GenMol generates novel scaffolds — filtered by RDKit, docked by DiffDock, ranked by evidence